

Improving Coordination in Teams: Combining Team and Individual Relative Performance Evaluations

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Abstract. Employees working in teams often fail to coordinate because of strategic uncertainty, i.e., difficulty predicting their colleagues' actions. Using an incentivized economic experiment, we examine the potentially beneficial effects of a ubiquitous competitive incentive—individual relative performance evaluations (RPE)—in reducing strategic uncertainty and improving team output. When employees work in highly interdependent teams and are motivated by team incentives, we predict that individual RPE reduce employees' strategic uncertainty and, in doing so, improve coordination. Consistent with our theory, we find that the use of individual RPE reduces employees' strategic uncertainty about both their team members' direct effort and indirect actions (i.e., helping or hurting). Furthermore, by reducing each component of strategic uncertainty, we find that firms can effectively improve coordination and team output. Contrary to claims that competitive, individual-level incentives harm team outcomes, our study highlights that individual RPE play an important role in reducing strategic uncertainty such that their use in interdependent teams motivated by cooperative team incentives yields superior outcomes.

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1. Introduction

Minimizing obstacles to coordination within teams is critical to firm success in the modern workplace. In contrast to groups of independent employees, teams are interdependent in nature such that the marginal effect of one employee's actions on team output depends on colleagues' actions (Katzenbach and Smith 1993, Weber 2000, Arnold and Tafov 2019). This interdependence makes coordination vital to team success, as coordination failures are incredibly costly to firms (Sims 2000, Brandts and Cooper 2006, Couto 2009). Such coordination failures often stem from strategic uncertainty, which is commonly defined in economics as the difficulty predicting how one's colleagues will act in team settings (Van Huyck et al. 1990). Our study considers two important and distinct components of strategic uncertainty in teams: strategic uncertainty about how much a team member will contribute (1) directly toward team output ("direct effort"), and (2) indirectly through helping or hurting ("indirect actions"). We investigate whether and how a pervasive form of incentive contracting—relative performance evaluations ("RPE")—can improve

coordination and team performance by reducing each component of strategic uncertainty.

RPE are prevalent in organizations (Hannan et al. 2008, Casas-Arce and Martínez-Jerez 2009). Team-level RPE ("team RPE"), where pay is based on a team's performance relative to that of other teams, are common and provide incentives for cooperative behaviors that support team performance (Arnold and Tafov 2019). Individual-level RPE ("individual RPE"), where pay is based on one's individual performance relative to that of peers, provide incentives for competitive behaviors that can limit cooperation within teams (Wang 2017). Nevertheless, individual RPE are ubiquitous in practice. Whereas some firms rely on individual RPE explicitly (e.g., through bonus plans or recognition awards), they are also universally used implicitly because of promotion and retention decisions. In this study, we predict that individual RPE reduce strategic uncertainty because they improve valuable coordination within teams—despite limiting cooperation—highlighting a key and novel benefit of the use of individual RPE in organizations.

Our research question is important for several reasons. Specifically, motivating team performance is a

growing concern because firms increasingly utilize teams to motivate their employees to coordinate (Harvard Business Review 2017, Shellenbarger 2018, Darino and Johnson 2020). As strategic uncertainty is a major obstacle to coordination (Van Huyck et al. 1990), it is important to understand how contractual incentives affect strategic uncertainty. Additionally, RPE can entail both cooperative and competitive incentives, depending on the reference group. Importantly, we differentiate coordination from cooperation in our study. Cooperation reflects effort that positively supports a team member working toward a shared benefit (Kosfeld and von Siemens 2011), whereas coordination only requires synchronized efforts and is not directional (i.e., neither positive nor negative) (Klein and Orsborn 2009). Competitive incentives (such as individual RPE) are thought to impair coordination and harm team outcomes, whereas team RPE are considered cooperative incentives that can improve coordination in team settings (Boss 2015, Arnold and Tafkov 2019). Given that individual RPE are ubiquitous, we contribute to the literature by examining an important alternative path by which individual RPE affect team outcomes. We predict that individual RPE reduce employees' strategic uncertainty about their team members and, in doing so, reduce costly coordination failures. Moreover, research typically focuses on the role of RPE in reducing common uncertainty affecting employee performance, thereby facilitating more efficient incentive contracting between the firm and its employees (Holmstrom 1982, Dye 1984). Our study provides distinct and novel insights into contracting efficiency in interdependent teams by being the first to examine how individual RPE reduce strategic uncertainty. In doing so, our study explains the ubiquitous use of individual RPE in practice (Parker et al. 2000, Van Bavel and Packer 2016, Hess 2023).

In all types of interdependent teams, employees must anticipate their team members' behaviors, when choosing their own costly direct effort and indirect actions, as the marginal return to their own behavior, and its effect on team output depends on their team members' behaviors. Therefore, coordination among team members is crucially important. Consistent with research in other settings, we expect that the competitive nature of individual RPE in interdependent teams will increase employees' direct effort in pursuit of RPE rewards (e.g., bonuses, promotions) and decrease the amount of cooperation (i.e., helping) among team members (Frederickson 1992, Barnes et al. 2011, Wang 2017). Importantly, we predict that employees anticipate these effects, which reduces their strategic uncertainty. We further predict that reduced strategic uncertainty improves coordination and, ultimately, team output as employees seek to coordinate with their team members by providing higher effort.

We test our hypotheses using an abstract laboratory experiment that incorporates a minimum-effort coordination game adapted to allow for indirect actions in addition to direct effort (Van Huyck et al. 1990). This setting entails a "weak-link" production function that captures the key theoretical features of interdependent teams (i.e., task interdependence among two homogeneous, effort-averse employees where effort is costly and strategic uncertainty exists). A weak-link team is clearly interdependent, as team output is driven by the lowest team member's output. Therefore, the return to one's behavior depends on whether it affects the lowest team member's output, which is a function of all other team members' behaviors. Given this strong form of interdependence, strategic uncertainty is a major deterrent to coordination and team output (Van Huyck et al. 1990). Moreover, weak-link production functions are common in practice, especially in firms that use sequential processes (e.g., construction, manufacturing, IT, airlines).¹ Our abstract setting also provides measures of direct effort, cooperation (i.e., indirect actions), strategic uncertainty, and coordination. Therefore, this setting allows us to cleanly separate the effects of individual RPE on these constructs when examining whether individual RPE improve coordination by reducing strategic uncertainty, despite potentially harming cooperation.

Within this experimental setting, we employ a 2×2 between-participants design, manipulating the presence (absent versus present) of two forms of RPE (team versus individual). This results in four conditions: *No RPE*, *Team RPE Only*, *Individual RPE Only*, and *Both RPE*. These conditions allow us to examine the incremental effect of the use of individual RPE when cooperative team-level incentives are relatively weak (*Individual RPE Only* versus *No RPE*) and strong (*Both RPE* versus *Team RPE Only*). Specifically, we randomly assigned participants to two-person teams, where they each chose levels of costly direct effort and indirect actions, earning a piece rate for team output, which was constrained by the lowest-performing team member (i.e., a "weak-link" production function). In addition, employees could also earn a bonus based on RPE, where present, with the reference group depending on the experimental condition. To cleanly test the distinct effect of individual RPE on strategic uncertainty, we hold the size of the bonus constant across RPE conditions and limit any reputational concerns or social norms related to teamwork.

Our results are consistent with our hypotheses. First, we find that individual RPE increase direct effort and decrease cooperation. Second, we find that individual RPE reduce employees' strategic uncertainty about their team members' direct effort, as well as reduce employees' strategic uncertainty about their team members' indirect actions when cooperative incentives are strong. Third, we document that reducing strategic

uncertainty improves coordination and team output; that is, when strategic uncertainty is reduced, employees seek to coordinate with their team members by providing higher direct effort. Importantly, our results highlight that individual RPE improve interdependent team output because they improve coordination, despite decreasing cooperation.

Our study makes several important contributions to the literature and practice. Reducing strategic uncertainty is a major issue for both practitioners and researchers, as employees often cannot observe their team members' actions before acting themselves (March and Simon 1958, Hamman et al. 2007, Starvish 2012, Schwarz 2017). Our findings highlight that individual RPE play a key role in reducing strategic uncertainty in interdependent teams motivated by team-level incentives. Specifically, we document how individual RPE can improve coordination and team output despite harming cooperation.

We also contribute to the literature on cooperation and coordination under strategic uncertainty (Knez and Camerer 1994, Klein and Orsborn 2009, Kosfeld and von Siemens 2011). To the best of our knowledge, our study is the first to explore the combination of team and individual incentives in interdependent teams, highlighting the value of individual incentives in such settings. Specifically, we provide novel evidence that intrateam competition can reduce strategic uncertainty in interdependent teams and, in doing so, can improve coordination and team output even while reducing cooperation. These results are important given the ubiquitous nature of individual RPE and their common critiques.

For example, business commentators often highlight the adverse effects of individual incentives in teams noting that “the practice of comparing and pitting workers against one another often hurts employee morale, collaboration, and productivity” (Hess 2023). Research also often focuses on the negative effects of individual incentives on team outcomes such as employees sabotaging or withholding help from team members (Arnold and Tafov 2019, Bloomfield et al. 2023). Nevertheless, despite these criticisms, individual incentives remain ubiquitous in practice. Recent press reports indicate that the ordinal/stacked ranking of employees for evaluation purposes is rising again, and almost one-third of Fortune 500 firms utilize such methods (Hess 2023). Bridging this disparity between research and practice, our study provides important insights into the positive effects of individual RPE by showing that firms can benefit from using competitive individual RPE in team settings where coordination is crucial. Furthermore, given that firms are often inextricably tied to individual incentives through performance-based promotions and dismissals or distribution of employee perks and responsibilities, our study suggests it can be useful for firms to emphasize

these individual incentives to improve coordination, rather than downplaying them.

Finally, we also contribute to the academic literature on reference groups for RPE (Aranda et al. 2014, Casas-Arce et al. 2018). Research typically examines either group or individual RPE in isolation (Matsumura and Shin 2006, Hannan et al. 2008, Tafov 2013, Arnold and Tafov 2019). Our study provides new insights into the complementary effects of the use of both forms of RPE. As such, we address an important gap in the literature by highlighting that individual RPE can mitigate the strategic uncertainty that persists in the presence of team RPE.

2. Theory Development and Hypotheses

2.1. Interdependent Teams and Strategic Uncertainty

Examining our research question necessitates a general setting with four key theoretical features: (1) interdependent teams of (2) effort-averse employees, where (3) effort is costly, and (4) strategic uncertainty exists. The term “group” broadly reflects a set of employees whose productive activities could be either independent or interdependent (Katzenbach and Smith 1993). In contrast, a distinguishing feature of “teams” is their interdependency where the relative effect of an employee's behaviors on team output depends on all other team members' behaviors (Morgan et al. 1986, Weber 2000).² Because of this interdependency, motivating teams presents a unique control problem (Baines and Langfield-Smith 2003, Pizzini 2010, Harvard Business Review 2017, Schwarz 2017, Shellenbarger 2018).

Interdependency requires employees to coordinate their actions. Coordination failures lead to wasted effort and inefficiencies (Wageman and Baker 1997, Wageman 2001). Assuming employees only want to engage in a costly behavior when its marginal benefit exceeds its marginal cost, employees working in interdependent teams must choose what behavior is best to engage in given the behaviors of their team members (Rankin 2004, Starvish 2012, Schwarz 2017). Nevertheless, when information is imperfect, imprecise, or untimely, it is difficult for employees to predict their team members' behaviors. Economics research refers to this difficulty as strategic uncertainty (Van Huyck et al. 1990). In a team with relatively homogenous employees, employees face strategic uncertainty about their team members' direct effort (i.e., costly effort directed toward their own performance) and indirect actions (i.e., costly effort directed toward other team members' performance). Strategic uncertainty leads to adverse team outcomes, such as dead-weight losses, inefficiencies, and achieving less Pareto-preferable equilibria (Knez and Camerer 1994, Ochs 1995, Cooper 1999, Brandts and Cooper 2006, Couto 2009).

Following research in economics on coordination and strategic uncertainty with interdependent actions, we model our economic setting using a minimum-effort coordination game with homogeneous employees who are each subject to the same incentive contract (Van Huyck et al. 1990, Knez and Camerer 1994, Weber 2000, Knez and Simester 2001, Hamman et al. 2007) and adapt this game to allow for employee cooperation.³ In this minimum-effort setting, team output is driven by the lowest team member's output. This setting affords us a clean test of theory, as it represents a strong form of interdependency with substantial strategic uncertainty. Furthermore, it presents a production function that is prominent both in economics (i.e., a Leontief (1936) or "weak-link" production function) and practice (e.g., construction, manufacturing, IT, airlines).⁴ We further discuss this production function in Section 3.1.

This setting also enables us to cleanly test how individual RPE affect three distinct constructs: (1) direct effort, (2) cooperation (i.e., indirect actions), and (3) coordination. Direct effort reflects employees' costly effort that directly affects their own output. Cooperation reflects effort that supports a team member when working toward a shared benefit (Kosfeld and von Siemens 2011).⁵ In interdependent teams, employees are weakly motivated to cooperate because of the nature of their interdependent efforts (i.e., to raise the minimum output within a team). Alternatively, coordination reflects employees working in sync to match efforts across team members (see Klein and Orsborn (2009) for a conceptual review of coordination). Coordination is not directional (i.e., positive or negative), as it only requires synchronized efforts.

Research on coordination in interdependent teams uses minimum-effort coordination games to examine a variety of factors impacting coordination, including team incentives (Bornstein et al. 2002), incomplete information (Asheim and Yoo 2008), and group size (Knez and Camerer 1994). However, to the best of our knowledge, no empirical studies explore the combination of team and individual incentives in interdependent teams. For example, whereas Riechmann and Weimann (2008) examine competition as a coordination device in a minimum-effort coordination game, they focus exclusively on interteam competition (i.e., team RPE) as a mechanism for increasing competition between teams to improve team output. In contrast, we examine intra-team competition. That is, we predict that the presence of competitive incentives within teams reduces strategic uncertainty and, in doing so, improves coordination and team output. We next develop our theory in detail.

2.2. Relative Performance Evaluations

Firms often use RPE to determine employees' pay by comparing the performance of a team or employee to a

reference group such as a peer team or peer employee (Aranda et al. 2014, Feichter et al. 2024). Research finds that RPE affect direct effort and productivity, as well as cooperation and competition depending on the reference group (Hannan et al. 2008, Hannan et al. 2013, Newman and Tafov 2014, Wang 2017, Casas-Arce et al. 2018).

Team RPE, which compare the performance of one team to another team's performance, are well suited to team settings, as they provide a cooperative incentive that motivates employees to improve team performance through cooperation (Pearsall et al. 2010, Boss 2015, Schwarz 2017). For example, Delfgaauw et al. (2013) highlight a large Dutch retail chain that uses team RPE to incentivize teams across stores. Furthermore, Bodell (2019) discusses how teams at 3M are tasked with developing improvements in energy efficiency, with the top team earning special rewards.

In contrast, individual RPE compare an employee's individual performance to that of other employees. Individual RPE reflect a competitive incentive that motivates employees to improve their own performance compared with others.⁶ Examples of individual RPE are ubiquitous in practice, both explicitly (e.g., tournaments, ranked bonus pool allocations, employee-of-the-month programs) and implicitly (e.g., retention/promotion). Even stacked ranking, once condemned for its competitive focus and negative impact on firm culture, is on the rise again, with about 30% of Fortune 500 firms adopting the practice (Allen 2012, Hess 2023). The ubiquitous use of individual RPE in practice seemingly contradicts the common argument that competitive incentives harm team performance (Rosenbaum et al. 1980, Pearsall et al. 2010, Barnes et al. 2011, Olson 2013, Boss 2015, Van Bavel and Packer 2016). As a step toward reconciling this argument with the ubiquitous use of individual RPE in practice, we now develop theory on how individual RPE affect employees' direct effort, indirect actions, strategic uncertainty, and coordination in teams.

2.3. Direct Effort

Providing high direct effort in interdependent teams is risky, as the relative effect of an employee's effort on team output depends on all other team members' actions. If an employee's effort goes unmatched, team performance may not sufficiently increase to compensate that costly effort. Absent common knowledge of each employee's preferences and rationality, unraveling can occur (Nagel 1995). For example, employees may hedge their direct effort to avoid the risk of providing more effort than is optimal in response to their team members' choices. If they believe that their team members will behave similarly, those employees must hedge even further, and so on. Therefore, strategic uncertainty about direct effort exists in interdependent

teams that likely leads risk-averse employees to limit their direct effort (Schwarz 2017).⁷

In general, research finds that the competitive nature of individual RPE increases direct effort as employees focus on improving their individual performance (Frederickson 1992, Fatas et al. 2006, Li et al. 2016). For example, Barnes et al. (2011) find that competitive incentives lead all team members to contribute higher direct effort than they would absent those incentives. Although some studies find that highly competitive individual RPE, such as tournaments, can induce complacency in top performers or giving-up behavior in bottom performers, this result is typically driven by widening skill gaps and heterogeneity among employees (Hannan et al. 2008, Casas-Arce and Martínez-Jerez 2009, Berger et al. 2013). In comparisons of homogenous employees, the motivating effect of individual RPE likely dominates, as employees are equally capable of obtaining the reward ex ante. By using individual RPE, a firm can effectively increase the expected value of an employee's direct effort even when that effort does not improve team performance. In this situation, unmatched direct effort still allows employees to increase their likelihood of receiving individual RPE rewards. Therefore, Hypothesis 1 predicts that the presence of individual RPE will increase employees' direct effort in a team setting.

Hypothesis 1. *In settings where employees work in highly interdependent teams and are motivated by team incentives, individual RPE increase direct effort.*

2.4. Indirect Actions

Indirect actions reflect additional efforts contributed by an employee toward helping or hurting a team member. Indirect actions can positively or negatively impact team output, depending on their direction. Research finds that team-based incentives such as team RPE incentivize cooperation within teams, as they increase the value of cooperative indirect actions (i.e., helping) (Wageman and Baker 1997, Chen and Li 2009, Kugler et al. 2010, Barnes et al. 2011, Arnold and Tafov 2019). In contrast, the competitive nature of individual RPE can reduce cooperation and motivate negative indirect actions (i.e., hurting), such as sabotage and withholding information (Quigley et al. 2007, Harbring and Irlenbusch 2011, Trikha 2012, Charness et al. 2014, Wang 2017). Consistent with these arguments, Hypothesis 2 predicts that the competitive effect of individual RPE on indirect actions will hold in our setting.

Hypothesis 2. *In settings where employees work in highly interdependent teams and are motivated by team incentives, individual RPE reduce cooperation (i.e., net indirect actions).*

2.5. Strategic Uncertainty

Our first two hypotheses related to direct effort (Hypothesis 1) and indirect actions (Hypothesis 2) provide a

foundation for developing our theory related to strategic uncertainty. Recall that strategic uncertainty reflects an employee's difficulty in predicting team members' behaviors (Van Huyck et al. 1990).

First, consider strategic uncertainty about direct effort. As previously discussed, individual RPE increase the expected value of an employee's direct effort even when that effort does not improve team performance. Hence, Hypothesis 1 predicts that individual RPE motivate higher direct effort by focusing employees on improving their individual performance (Fatas et al. 2006, Li et al. 2016). To predict how individual RPE affect employees' strategic uncertainty about direct effort, we require several assumptions. Specifically, as described in the basic economic setting (see Section 2.1), we assume that all employees are homogeneous and are subject to the same incentive contract. We also assume that employees are boundedly rational (Selten 1990) and are prone to common heuristics and biases, such as the "false consensus" or "assumed similarity" bias (Cronbach 1955, Ross et al. 1977, Mullen et al. 1985). That is, individuals often expect other people to react similarly to themselves. In doing so, they overestimate the commonness of their own behavior, and ultimately, this bias leads to a "false consensus" (Ross et al. 1977, Mullen et al. 1985).

Based on these assumptions, if employees expect that their team members will react to individual RPE similarly to themselves by also contributing higher amounts of direct effort, then strategic uncertainty about such direct effort should decrease. Importantly, individual RPE provide employees with a directional prediction when they develop their expectations regarding their team member's effort. We therefore assume that employees will update their beliefs about the relative likelihood of higher direct effort in the presence of individual RPE, thereby reducing the expected range of their team member's behavior and decreasing employees' strategic uncertainty about direct effort. This leads to the following directional prediction:

Hypothesis 3a. *In settings where employees work in highly interdependent teams and are motivated by team incentives, individual RPE reduce employees' strategic uncertainty about their team members' direct effort.*

Now, consider strategic uncertainty about indirect actions. As previously established, employees in teams are weakly motivated to cooperate because of the nature of their interdependent efforts (i.e., to raise the minimum effort within the team). When firms contract on team RPE, the incentive to cooperate increases, as it raises the value of cooperative indirect actions (Wageman and Baker 1997, Kugler et al. 2010, Arnold and Tafov 2019). In contrast, the competitive nature of individual RPE motivates the opposite behavior (i.e., negative indirect actions or reduced cooperation)

(Harbring and Irlenbusch 2011, Charness et al. 2014). As such, individual RPE introduce a countervailing directional incentive for indirect actions in teams that may ultimately reduce employees' strategic uncertainty about their team members' indirect actions.

Specifically, when employees are strongly motivated to cooperate in teams, we expect that the countervailing directional effect of individual RPE reduces the expected likelihood and magnitude of indirect actions (either positive or negative). For example, assume the relative magnitude of each opposing incentive is equal. In that case, individual RPE weaken the motivation to cooperate that stems from team incentives by inducing competition for individual rewards (Charness et al. 2014), whereas team incentives weaken the motivation to hurt stemming from individual RPE by increasing the value of cooperation (Wageman and Baker 1997). In this case, we predict that reducing the absolute value of any indirect actions would also reduce employees' expectations regarding the absolute value of their team members' indirect actions, effectively reducing employees' strategic uncertainty about such actions.

However, there is important tension underlying this directional prediction. As our study is the first to examine the effects of these simultaneous countervailing incentives, there is no assurance that the relative magnitude of these effects will be equal. That is, although individual RPE weaken the motivation to cooperate and incentivize negative indirect actions (Harbring and Irlenbusch 2011, Charness et al. 2014), it is unclear whether such a strong competitive incentive will partially, fully, or more than fully offset the desire to cooperate in interdependent teams that are strongly motivated to cooperate. Moreover, this tension is even greater when there are only weak incentives to cooperate (i.e., absent explicit team RPE). As such, the use of individual RPE in isolation may not decrease (and may even increase) employees' strategic uncertainty about their team members' indirect actions relative to a setting with no RPE. Given this inability to predict the relative magnitude of these countervailing effects *ex ante*, we state Hypothesis 3b in null form:

Hypothesis 3b. *In settings where employees work in highly interdependent teams and are motivated by team incentives, individual RPE have no effect on employees' strategic uncertainty about their team members' net indirect actions.*

2.6. Team Output

We now develop theory involving team output as a function of strategic uncertainty. In settings absent strategic uncertainty, individual RPE have been shown to increase output through improved direct effort (Barnes et al. 2011, Li et al. 2016), but also harm output through less cooperation (Wang 2017). In settings involving

strategic uncertainty, individual RPE not only motivate these effects on direct effort (Hypothesis 1) and cooperation (Hypothesis 2) but also reduce employees' strategic uncertainty about their team members' direct effort (Hypothesis 3a) and, possibly, indirect actions (Hypothesis 3b). By definition, reducing strategic uncertainty should improve coordination, as employees are better able to predict the actions of their team members. Collectively, these arguments suggest that individual RPE will enable greater coordination, facilitating higher levels of coordinated direct effort and thereby improving team output. Importantly, this improvement in coordination is distinct from simply increasing the intensity of a direct effort incentive. Rather, reducing employees' strategic uncertainty about their team members' behaviors makes coordinating effort easier and increases team output. This leads to our fourth hypothesis, stated separately for strategic uncertainty about direct effort (Hypothesis 4a) and net indirect actions (Hypothesis 4b):

Hypothesis 4a. *In settings where employees work in highly interdependent teams and are motivated by team incentives, reducing employees' strategic uncertainty about their team members' direct effort increases team output.*

Hypothesis 4b. *In settings where employees work in highly interdependent teams and are motivated by team incentives, reducing employees' strategic uncertainty about their team members' net indirect actions increases team output.*

Together, Hypothesis 3 and Hypothesis 4 imply that individual RPE improve team output. That is, a reduction in strategic uncertainty should mediate a positive effect of individual RPE on team output. Importantly, there is tension regarding whether the overall effect of individual RPE on team output is positive. Specifically, Hypothesis 2 highlights that individual RPE motivate less cooperation and potentially induce negative indirect actions (Harbring and Irlenbusch 2011, Charness et al. 2014). However, if the benefits of improved coordination at higher levels of direct effort—mediated by reduced strategic uncertainty—outweigh the cost of less cooperation, individual RPE would improve team output overall. We empirically test the overall effect of individual RPE and this potential mediation through strategic uncertainty in Section 4.4.

3. Experimental Method and Design

3.1. Economic Setting

We test our theory using an abstract laboratory experiment that incorporates a two-person minimum-effort coordination, or “weak-link,” game (Van Huyck et al. 1990). This experimental setting affords us a clean test of theory and captures four key theoretical features: (1) interdependent teams of (2) effort-averse employees,

where (3) effort is costly and (4) strategic uncertainty exists. Moreover, it allows us to isolate the distinct effects of individual RPE on direct effort, indirect actions, and coordination within teams.

In our setting, there are two homogeneous employees per team. Employees earn a piece rate based on team output. As each employee's pay is contingent on team output, it reflects a team-based incentive encouraging cooperation. Each employee provides costly direct effort that maps into his or her own output, and can also take costly indirect actions that increase or decrease the team member's output. To ensure that strategic uncertainty exists, employees make both choices before learning their team member's choices.

The production function is such that each unit of team output requires one unit of output from each employee within the team. Therefore, team output is equal to the lowest team member's output, ensuring that our team setting is interdependent. Specifically, the direct effort and indirect actions of a team member can influence the marginal effect of an employee's own choices on team output, and vice versa.⁸ We set the cost of either direct effort or indirect actions at less than the piece rate for team output such that any strategy where output is coordinated is a Nash equilibrium (Van Huyck et al. 1990). This results in multiple Pareto-ranked equilibria where coordination is valuable and strategic uncertainty about which equilibrium a team member is pursuing restricts team output. These features lead to a profit function for each employee of:

$$\text{profit} = (\text{piece} - \text{rate} \times \text{team output}) - \text{cost}.$$

Incremental to this basic design, employees can also earn a bonus based on RPE in the conditions of interest, with the RPE determined by the experimental condition to which they are assigned.⁹ We set the bonus at a sufficiently large amount to offset most (if not all) of the cost of greater direct effort or indirect actions in pursuit of the bonus, discussed further in Section 3.4. This design feature ensures that the bonus is salient, effective, and reflects the costs and benefits of any helping or hurting to earn these bonuses.

3.2. Experimental Design

To test our hypotheses, we employ a 2×2 between-participants experimental design where we manipulate the use of two forms of RPE (team versus individual) at two levels (absent versus present). This design allows us to test the effects of individual RPE when team incentives are relatively weak (team RPE absent) and relatively strong (team RPE present). We refer to the four conditions as *No RPE*, *Team RPE Only*, *Individual RPE Only*, and *Both RPE*.

In each condition, participants are randomly matched into two-person teams to complete a two-stage chosen effort task that repeats over 15 periods. To control for

norm development and reciprocal punishment, we randomly rematch participants with a new team member each period such that no two participants are ever in the same team twice. We discuss how this design choice facilitates a clean test of theory in further detail in Section 3.5. We ran two sessions per condition (eight in total), with a total of 128 undergraduate participants recruited from a large U.S. university. The condition for each session was randomly determined. We conduct our experiment using z-Tree (Fischbacher 2007). On average, participants were 20.7 years old with 22.6 months of work experience, and 48% were female. Each session took around 90 minutes, and participants' average pay was \$18.32, including a \$5 show-up fee.

3.3. Task Details and Procedures

Upon entering the laboratory, participants gave their informed consent and were randomly assigned to a private computer terminal.¹⁰ Participants then read instructions for their condition. All participants within a session were in the same condition. After the instructions were read aloud by one of the researchers, participants were required to correctly complete a manipulation check quiz before continuing. Then, before the 15 live periods, participants completed five practice periods paired with the computer. In these practice periods, participants were informed that the computer randomly chose its direct effort and indirect actions level. These practice periods served to ensure participants understood their action set, payoffs, and the interface.

In each period, participants completed a two-stage chosen effort task. Participants chose a level of direct effort that corresponded to their own output and a level of indirect actions that increased or decreased their team member's output. While each participant paid for his or her own costly direct effort and indirect actions, team output was the lower of the two team members' output. Direct effort (indirect actions) mapped into a participant's (the team member's) individual output without noise to control for any concerns related to the realization of intended actions. A unit of team output was worth \$1 to each team member, whereas a unit of direct effort or indirect actions cost \$0.50. As the piece rate per unit of team output was larger than the unit cost of direct effort or indirect actions, any action that increased team output was optimal, and any action that did not increase team output was suboptimal (ignoring the RPE bonus).

In stage 1 of the two-stage chosen effort task, participants chose their direct effort level from the range [20,30]. In stage 2, without any new information, participants submitted a prediction of their team member's direct effort and chose whether, and by how much, to increase or decrease their team member's output (i.e., indirect actions). To encourage thoughtful predictions, participants could earn a \$1 bonus for correctly

predicting their team member's direct effort. We constrained the change to a team member's output to $[-10,10]$, but a participant could not force a team member's output outside of $[20,30]$. After stage two, participants learned the period results, including (1) their own output, (2) their team member's direct effort, (3) their team member's output, and (4) the output of a matched peer team in all conditions. Participants were then rematched with a new team member and started the next period. Table A.1 in Appendix A presents the payoff table.¹¹

Participants repeated this task across all 15 live periods, allowing us to check for consistency in choices. After the postexperimental questionnaire, the computer randomly selected one of the 15 live periods for payment. This payment approach incentivizes participants to treat each period independently and removes any possible wealth effects (Croson 2005).

3.4. Manipulations

We manipulated between-participants the presence of two forms of RPE: team RPE and individual RPE. The *No RPE* condition did not have any possible bonus available. The aggregate size of the bonus(es) available to each participant was constant across all three RPE conditions at \$6. We set the bonus amount at \$6 to offset most (if not all) of the marginal cost of effort in pursuit of the bonus, ensuring the bonus is sufficiently lucrative without inducing ceiling effects.

For example, consider the individual RPE bonus. As the marginal cost of direct effort is \$0.50 per unit and there are 10 possible unit levels, the maximum differential along any given minimum output is \$5. Setting the individual RPE bonus at \$6 fully offsets this additional cost. Moreover, when direct effort is initially matched, negative indirect actions effectively cost the participant \$1.50 per unit (including the reduction in base pay) but may allow them to earn the individual RPE bonus, again offsetting this cost. These examples highlight scenarios where the \$6 bonus potentially offsets all of the marginal cost of effort when a team member's actions are known with certainty, whereas it may only partially offset costs depending on the simultaneous actions of the other team member because of the interdependent nature of this setting.

All RPE were based on a paired comparison to hold constant the level of competition. The *Team RPE Only* condition randomly paired two teams each period and compared the output of these teams. Each team member on the team with the higher team output earned a \$6 bonus. The *Individual RPE Only* condition compared the individual output of the two team members within the same team each period. The team member with the highest individual output within each team earned a \$6 bonus. The computer randomly determined the bonus in cases with ties between pairs.¹²

In the *Both RPE* condition, both team and individual output were compared. Therefore, to determine bonuses, we compared (1) two randomly paired teams, and (2) the two team members within a given team. Both team members on the higher performing team earned \$3 (i.e., the team RPE bonus), and the highest performing team member within each team earned \$3 (i.e., the individual RPE bonus).

3.5. Additional Design Features

To test whether individual RPE can effectively reduce strategic uncertainty, we ensure our setting involves high strategic uncertainty in the baseline condition. Knez and Camerer (1994) argue that such a design is akin to having a reliable way to induce illness in patients to examine the efficacy of a cure for that illness. Accordingly, participants make their choices simultaneously, without prechoice communication or commitment, and interact in an anonymous one-shot setting with random rematching between periods. These requirements increase strategic uncertainty and facilitate our test of theory.

Moreover, although communication and reputation often exist in workplace environments, strategic uncertainty persists in practice (Couto 2009, Starvish 2012, Boss 2015, Schwarz 2017). Factors such as noisy production functions, remote work, employee turnover, changing teams, temporal and spatial distance between team members, fluctuating timelines, and cheap talk contribute to strategic uncertainty, despite communication among team members. Therefore, while communication and reputation can improve coordination in laboratory settings (Devetag and Ortmann 2007, Cooper and Weber 2020), these features are insufficient to eliminate strategic uncertainty in practice. As the focus of this study is to test whether individual RPE can reduce strategic uncertainty in interdependent teams, we prioritize establishing an experimental setting where high strategic uncertainty exists to aid our test of theory.

Finally, it is important that participants understand the implications of their choices in our setting. Therefore, after 10 live periods, participants received the following on-screen alert:

In order to provide clarification on your choices, the last 5 periods of this experiment will include a few additional questions. Nothing else about the experiment has changed. All payoffs in the matrix and instructions remain consistent with the prior periods.

Recall that in stage 2, after we elicited participants' predictions of their team member's direct effort, participants indicated by how much they would like to change their team member's output, if at all. To do so, they entered a change from $[-10,10]$, with zero indicating no change. During periods 11–15 (i.e., the

“guided-language” periods), we first explicitly asked participants whether they wanted to increase, decrease, or neither increase nor decrease the likelihood that their team member’s output and/or their peer team’s output was greater than their own before entering the size of the change. If they selected not to make any change, we asked them to confirm their decision.¹³ The purpose of these guided-language periods was to ensure that our measures captured intentional behavior. We find no significant differences in our results between the nonguided and guided-language periods in untabulated analyses. Therefore, it appears that participants properly understood the implications of their choices throughout the experiment.¹⁴

3.6. Variable Descriptions

Table B.1 in Appendix B describes our variables. Our dependent variables related to strategic uncertainty and team output are constructed using several measures. Each period, participants chose their level of *Direct Effort* and *Net Indirect Actions*. *Direct Effort* is a participant’s direct effort in the range [20,30]. *Net Indirect Actions* is how much a participant chose to change his or her team member’s output in the range [−10,10]. As that change could reflect either positive or negative indirect actions, we separately define *Helping (Hurting)* as the absolute value of that change if it is greater than (less than) zero, and zero otherwise. We measure *Team Output* as the lower of the team members’ individual outputs, as disclosed to participants.

As a general proxy for overall strategic uncertainty, we measure *Coordination Failure* as the absolute value of the difference between team members’ output in a given period, ranging from [0,10]. However, it is important to measure the distinct components of strategic uncertainty to test our theory. Therefore, we measure both strategic uncertainty about a team member’s direct effort (*Uncertainty_Direct*) and strategic uncertainty about a team member’s indirect actions (*Uncertainty_Indirect*). *Uncertainty_Direct* is the absolute value of the difference between participant i ’s incentivized prediction of team member j ’s direct effort ($E[e_j]$) and j ’s actual direct effort (e_j) (i.e., $|E[e_j] - e_j|$). *Uncertainty_Indirect* is equal to the absolute value of the difference between participant i ’s implicit expectation of team member j ’s indirect actions ($E[\delta_j]$) and j ’s actual indirect actions (δ_j) (i.e., $|E[\delta_j] - \delta_j|$).

To measure participant i ’s implicit expectation of team member j ’s indirect actions, we first calculate the participant’s expectation of j ’s output ($E[y_j]$) as the expected level of direct effort from j ($E[e_j]$) plus participant i ’s own chosen indirect actions (δ_i). In equilibrium, we assume that participants in the *Team RPE Only* and *No RPE (Both RPE and Individual RPE Only)* conditions attempt to match (produce one more output than) their team member’s output.¹⁵ In the *Team RPE Only* and

No RPE conditions, we calculate participants’ implicitly expected change as the difference between the expected output of their team member ($E[y_j]$) and their own direct effort choice (i.e., $E[y_j] - e_i$). In the *Both RPE* and *Individual RPE Only* conditions, we add one to this amount (i.e., $E[y_j] - e_i + 1$).¹⁶

Our independent variables of interest are indicator variables representing the experimental conditions. In addition, we elicit several potential covariates in a post-experimental questionnaire, including general demographic information and perceptions of fairness, risk aversion (Holt and Laury 2002), and social value orientation (Van Lange et al. 1997). To validate the effectiveness of our random assignment, we perform several untabulated tests confirming that there are no significant differences in participants’ demographics or perceptions of fairness across conditions. However, we do observe differences in participants’ risk aversion and social value orientation; however, our results are robust to including these control variables. Please refer to our Online Appendix for a comprehensive set of robustness tests. Our results are also robust to controlling for several other potential covariates, including prior period choices.

4. Results

Table 1 presents our descriptive statistics by condition. When testing our hypotheses, we use two-sided censored Tobit regressions when analyzing any dependent variable with mechanical limits. These dependent variable ranges are [0,10] for *Uncertainty_Direct*, *Helping*, *Hurting*, and *Coordination Failure*; [0,20] for *Uncertainty_Indirect*; and [20,30] for *Direct Effort* and *Team Output*.¹⁷ All of our regressions also include period fixed effects and cluster robust standard errors at the participant level.

4.1. Tests of Hypothesis 1: Direct Effort

Our first hypothesis (Hypothesis 1) predicts that the use of individual RPE will increase employees’ direct effort in settings with interdependent teams motivated by team incentives. Therefore, we focus our tests of Hypothesis 1 on the effects of individual RPE on *Direct Effort*, highlighting comparisons between (1) *Individual RPE Only* versus *No RPE*, and (2) *Both RPE* versus *Team RPE Only*. The pattern of results documented in panel A of Table 2 provides univariate support for our hypothesis; that is, *Direct Effort* is significantly higher in the *Individual RPE Only* condition compared with the *No RPE* condition (difference = 4.15, $t = 8.13$, $p < 0.01$) and in the *Both RPE* condition compared with the *Team RPE Only* condition (difference = 1.79, $t = 3.16$, $p < 0.01$).¹⁸ These results suggest that employees are less concerned that their direct effort will be unmatched

Table 1. Descriptive Statistics

	No RPE	Team RPE Only	Individual RPE Only	Both RPE	Overall
Age	20.66 (1.68)	20.69 (1.47)	21.06 (3.65)	20.22 (1.60)	20.66 (2.27)
Gender	1.56 (0.50)	1.56 (0.50)	1.41 (0.50)	1.38 (0.49)	1.48 (0.50)
Direct Effort	23.68 (3.46)	25.74 (3.22)	27.83 (3.19)	27.52 (2.60)	26.19 (3.54)
Net Indirect Actions	0.66 (1.48)	1.37 (1.73)	−0.97 (2.15)	0.06 (1.23)	0.28 (1.89)
Uncertainty_Direct	3.88 (2.98)	3.64 (2.62)	3.12 (3.16)	2.57 (2.30)	3.30 (2.83)
Uncertainty_Indirect	2.98 (2.99)	2.98 (2.63)	3.20 (3.07)	2.19 (2.17)	2.84 (2.76)
Coordination Failure	4.03 (3.00)	3.82 (2.99)	3.59 (3.04)	2.53 (2.49)	3.49 (2.94)
Team Output	22.22 (2.54)	24.87 (2.97)	25.00 (3.07)	26.20 (2.76)	24.57 (3.19)

Notes. This table reports descriptive statistics by condition and overall. We report the means (standard deviations) for Age and Gender at the participant level (i.e., $n = 32$ per condition) that reflect participant-level demographic information. We report all other variables at the participant-period level (i.e., $n = 480$ per condition). *No RPE* reflects the condition where no bonus is paid related to any relative performance. *Team RPE Only* reflects the condition where a bonus is paid to the team with the higher team output, comparing each team to only one other team. *Individual RPE Only* reflects the condition where a bonus is paid to the individual team member who produced the highest individual output in the team. *Both RPE* reflects the condition where bonuses are paid to both the team with the higher team output, comparing each team to only one other team, and the individual team member who produced the highest individual output in the team.

when individual RPE are used, both with or without team RPE. Interestingly, although not hypothesized, we also note that *Direct Effort* is not significantly different between the *Both RPE* and *Individual RPE Only* conditions ($p = 0.50$), indicating that the two conditions with individual RPE motivate similar levels of direct effort.

Panels B and C of Table 2 provide multivariate support for Hypothesis 1. For completeness, panel B reports the results of a Tobit regression using our full sample, with *Direct Effort* as the dependent variable and indicator variables for *Team RPE*, *Individual RPE*, and their interaction as independent variables. We observe a significant positive effect of *Individual RPE* on *Direct Effort* (coefficient = 6.70, $t = 7.59$, $p < 0.01$).¹⁹ To formally test Hypothesis 1, we then examine the results of our simple effects for individual RPE in panel C, where we control for period fixed effects and clustered robust standard errors. Consistent with Hypothesis 1, this test documents that individual RPE have a positive and significant effect on *Direct Effort* whether team RPE are absent (coefficient = 7.56, $t = 7.10$, $p < 0.01$) or present (coefficient = 2.37, $t = 3.08$, $p < 0.01$).

4.2. Tests of Hypothesis 2: Net Indirect Actions

Our second hypothesis (Hypothesis 2) predicts that the use of individual RPE will reduce cooperation among employees motivated by team incentives. Specifically, we expect to observe a reduction in net indirect actions when individual RPE are present in a setting where positive indirect actions reflect more cooperation (i.e.,

helping), and negative indirect actions reflect less cooperation (i.e., hurting). We find univariate support for this hypothesis in panel A of Table 3, as we observe that *Net Indirect Actions* is significantly lower in the *Individual RPE Only* condition compared with the *No RPE* condition (difference = -1.62 , $t = -6.39$, $p < 0.01$) and in the *Both RPE* condition compared with the *Team RPE Only* condition (difference = -1.31 , $t = -5.04$, $p < 0.01$). Panels B and C provide consistent multivariate support by showing that individual RPE have a negative and significant effect on *Net Indirect Actions* whether team RPE are absent (coefficient = -1.62 , $t = -6.44$, $p < 0.01$) or present (coefficient = -1.31 , $t = -5.08$, $p < 0.01$).

Exploring these effects further, Figure 1 presents the frequency and extent of *Net Indirect Actions* by condition. Consistent with findings from prior research, we find that cooperative indirect actions (i.e., *Helping*) are significantly higher in the *Team RPE Only* condition than in the *No RPE* condition (coefficient = 1.79, $t = 3.29$, $p < 0.01$; untabulated), and negative indirect actions (i.e., *Hurting*) are significantly higher in the *Individual RPE Only* condition than in the *No RPE* condition (coefficient = 4.63, $t = 8.42$, $p < 0.01$; untabulated). Interestingly, Figure 1 also highlights that the simultaneous use of both team and individual RPE reduces positive (negative) indirect actions relative to using team RPE (individual RPE) in isolation, making indirect actions less frequent and more muted when both forms of RPE are present. We confirm these effects are significant

Table 2. Effects of the Form of RPE on *Direct Effort*

Panel A: Mean comparisons, univariate					
	<i>No RPE</i> (1)	<i>Team RPE Only</i> (2)	<i>Individual RPE Only</i> (3)	<i>Both RPE</i> (4)	
Mean (standard deviation)	23.68 (3.46)	25.74 (3.22)	27.83 (3.19)	27.52 (2.60)	
	Prediction	Difference	Standard error	<i>t</i> -statistic	<i>p</i> -value
Hypothesized comparisons					
(3) versus (1)	+ (Hypothesis 1)	4.15	0.51	8.13	<0.01
(4) versus (2)	+ (Hypothesis 1)	1.79	0.56	3.16	<0.01
Other comparisons					
(2) versus (1)		2.05	0.61	3.38	<0.01
(4) versus (1)		3.84	0.51	7.60	<0.01
(3) versus (2)		2.10	0.57	3.68	<0.01
(4) versus (3)		−0.31	0.46	−0.68	0.50
Panel B: Full sample, multivariate					
		Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
<i>Team RPE</i>		3.01 (0.89)	3.39	<0.01	
<i>Individual RPE</i>		6.70 (0.88)	7.59	<0.01	
<i>Team RPE</i> × <i>Individual RPE</i>		−4.15 (1.20)	−3.46	<0.01	
Intercept		22.03 (0.70)	31.37	<0.01	
<i>n</i> = 1,920 (128 clusters), <i>F</i> (17, 1,903) = 5.37					
Panel C: Simple effects, multivariate					
	Prediction	Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> absent	+ (Hypothesis 1)	7.56 (1.06)	7.10	<0.01	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> present	+ (Hypothesis 1)	2.37 (0.77)	3.08	<0.01	

Notes. This table reports results related to *Direct Effort* that reflects participants' own direct effort choices between the possible range [20,30]. All *p*-values reflect two-tailed tests, and all variables reflect those defined in Table B.1 in Appendix B. Panel A reports the univariate results of a series of *t*-tests of mean differences across conditions with unequal variances at the participant level (*n* = 32 per condition). Conditions are labeled (1) *No RPE*, (2) *Team RPE Only*, (3) *Individual RPE Only*, and (4) *Both RPE*. Hypothesized comparisons are annotated with directional predictions, and all other comparisons are reported for completeness. Panels B and C report the results of multivariate Tobit regressions with *Direct Effort* as the dependent variable. Each model includes period fixed effects and clusters robust standard errors at the participant level. Panel B includes the full sample (*n* = 1,920), with 247 left-censored and 542 right-censored observations. All results are robust to using OLS as detailed in the Online Appendix. Panel C reports the hypothesized simple effects from these multivariate tests.

in untabulated tests, noting that *Both RPE* results in a significant decrease in *Helping* compared with the *Team RPE Only* condition (coefficient = −2.48, *t* = −5.29, *p* < 0.01) and a significant decrease in *Hurting* compared with the *Individual RPE Only* condition (coefficient = −2.48, *t* = −4.66, *p* < 0.01). We also find consistent results using the absolute value of these measures (i.e., *Net Indirect Actions*).²⁰

These results suggest that firms should consider the relative importance of direct effort and indirect actions within their objective function when designing RPE incentives. While it may be critical for some firms to improve direct effort and/or minimize negative indirect actions, other firms may rely heavily upon positive indirect actions within their organizational structures. Our results suggest the use of individual RPE may be useful for the former set of firms but detrimental to the latter set of firms.

4.3. Tests of Hypothesis 3a and Hypothesis 3b: Strategic Uncertainty

We now test our third hypothesis regarding the effects of individual RPE on strategic uncertainty about direct effort (Hypothesis 3a) and indirect actions (Hypothesis 3b). Before formally testing each component separately, we examine the effects of individual RPE on overall strategic uncertainty (i.e., *Coordination Failure*). In untabulated Tobit regressions, we find that individual RPE have a significant negative effect on *Coordination Failure*, indicating that coordination is better in the presence of individual RPE, whether team RPE are absent (coefficient = −0.50, *t* = −1.77, *p* = 0.08) or present (coefficient = −1.50, *t* = −3.89, *p* < 0.01). As illustrated in Figure 2, we also observe an improvement in the rates of full efficiency (i.e., perfect coordination) when individual RPE are present. Specifically, although there is not a significant improvement in

Table 3. Effects of the Form of RPE on Cooperation (*Net Indirect Actions*)

Panel A: Mean comparisons, univariate					
	No RPE (1)	Team RPE Only (2)	Individual RPE Only (3)	Both RPE (4)	
Mean (standard deviation)	0.66 (1.48)	1.37 (1.73)	−0.97 (2.15)	0.06 (1.23)	
	Prediction	Difference	Standard error	<i>t</i> -statistic	<i>p</i> -value
Hypothesized comparisons					
(3) versus (1)	− (Hypothesis 2)	−1.62	0.25	−6.39	<0.01
(4) versus (2)	− (Hypothesis 2)	−1.31	0.26	−5.04	<0.01
Other comparisons					
(2) versus (1)		0.71	0.27	2.59	0.01
(4) versus (1)		−0.60	0.19	−3.21	<0.01
(3) versus (2)		−2.33	0.31	−7.49	<0.01
(4) versus (3)		1.02	0.24	4.30	<0.01
Panel B: Full sample, multivariate					
		Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
Team RPE		0.71 (0.27)	2.62	<0.01	
Individual RPE		−1.62 (0.25)	−6.47	<0.01	
Team RPE × Individual RPE		0.31 (0.36)	0.87	0.39	
Intercept		0.95 (0.20)	4.70	<0.01	
<i>n</i> = 1,920 (128 clusters), <i>F</i> (17, 1,903) = 7.81					
Panel C: Simple effects, multivariate					
	Prediction	Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
Effect of Individual RPE if Team RPE absent	− (Hypothesis 2)	−1.62 (0.25)	−6.44	<0.01	
Effect of Individual RPE if Team RPE present	−(Hypothesis 2)	−1.31 (0.26)	−5.08	<0.01	

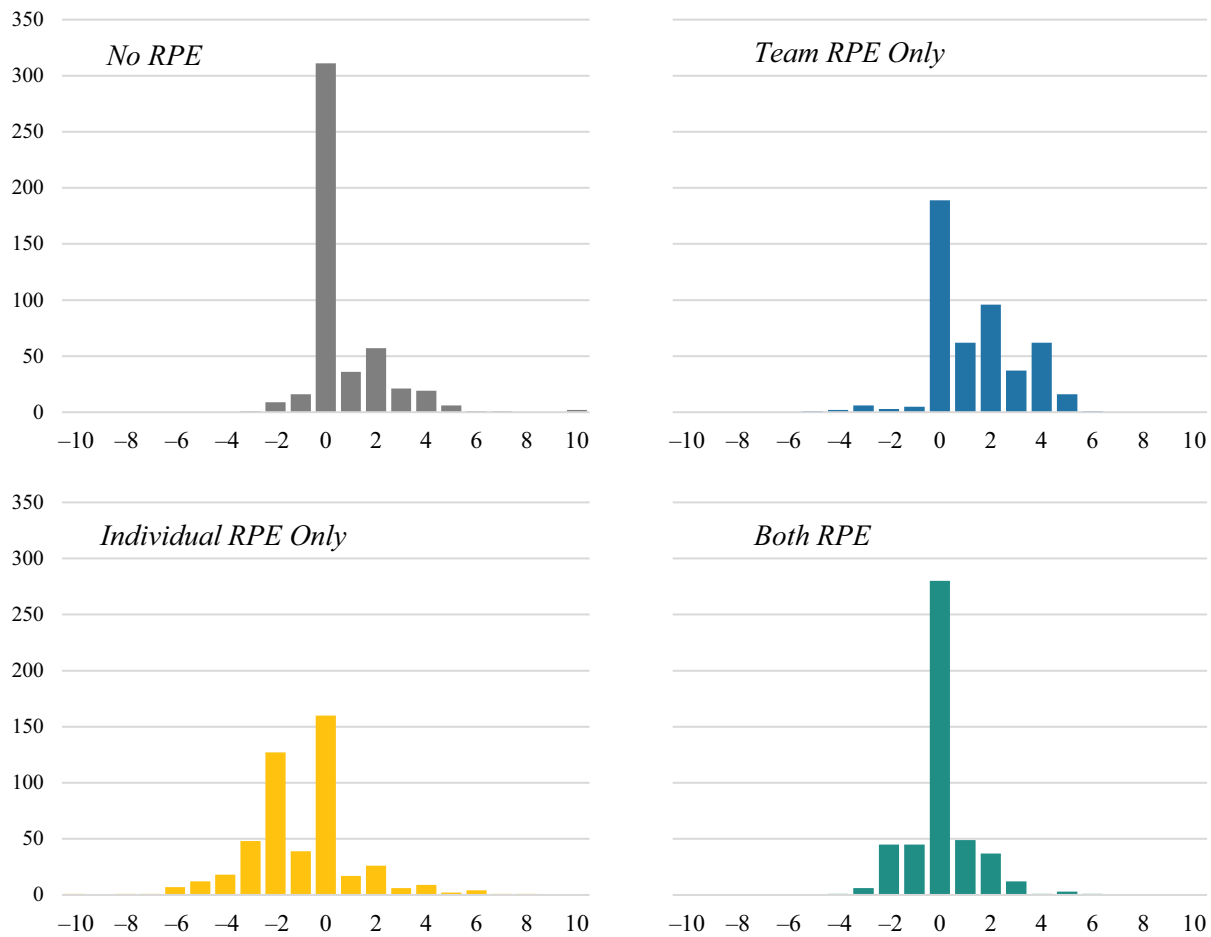
Notes. This table reports results related to *Net Indirect Actions* that reflect the amount by which each participant chose to change their team member's output within the range [−10,10]. All *p*-values reflect two-tailed tests, and all variables reflect those defined in Table B.1 in Appendix B. Panel A reports the univariate results of a series of *t*-tests of mean differences across conditions with unequal variances at the participant level (*n* = 32 per condition). Conditions are labeled (1) *No RPE*, (2) *Team RPE Only*, (3) *Individual RPE Only*, and (4) *Both RPE*. Hypothesized comparisons are annotated with directional predictions, and all other comparisons are reported for completeness. Panels B and C report the results of multivariate Tobit regressions with *Net Indirect Actions* as the dependent variable. Each model includes period fixed effects and clusters robust standard errors at the participant level. Panel B includes the full sample (*n* = 1,920), with one left-censored and two right-censored observations. All results are robust to using OLS as detailed in the Online Appendix. Panel C reports the hypothesized simple effects from these multivariate tests.

perfect coordination between the *Individual RPE Only* and *No RPE* conditions (difference = 0.02, *t* = 0.79, *p* = 0.43; untabulated), there is a significant increase in the *Both RPE* condition compared with the *Team RPE Only* condition (difference = 0.07, *t* = 2.70, *p* < 0.01; untabulated).²¹ Interestingly, participants in the *Both RPE* condition were able to perfectly coordinate 21.25% of the time, which is significantly higher than any other condition (all *p* < 0.01). Moreover, they were able to coordinate within two units of output in 60% of our observations, which is significantly better (all *p* < 0.01) than any of the other conditions (*No RPE*: 40%; *Team RPE Only*: 42%; *Individual RPE Only*: 50%). These results provide initial evidence that the use of individual RPE decreases overall strategic uncertainty and that incorporating individual RPE when cooperative incentives are strong (i.e., team RPE are present) is

particularly effective, as it leads to the lowest coordination failure overall.

We now examine the effects of individual RPE on each distinct component of strategic uncertainty. Hypothesis 3a predicts that strategic uncertainty about direct effort will be lower when individual RPE are present. The pattern of results in panel A of Table 4 is consistent with this hypothesis, as *Uncertainty_Direct* is lower in the *Individual RPE Only* condition compared with the *No RPE* condition (difference = −0.76, *t* = −2.81, *p* < 0.01) and in the *Both RPE* condition compared with the *Team RPE Only* condition (difference = −1.07, *t* = −4.41, *p* < 0.01). Panels B and C confirm these results using multivariate tests. In panel C, we observe a significant negative effect of individual RPE on *Uncertainty_Direct* when team RPE are absent (coefficient = −1.11, *t* = −2.97, *p* < 0.01) and when they

Figure 1. (Color online) Frequency of *Net Indirect Actions* by Condition



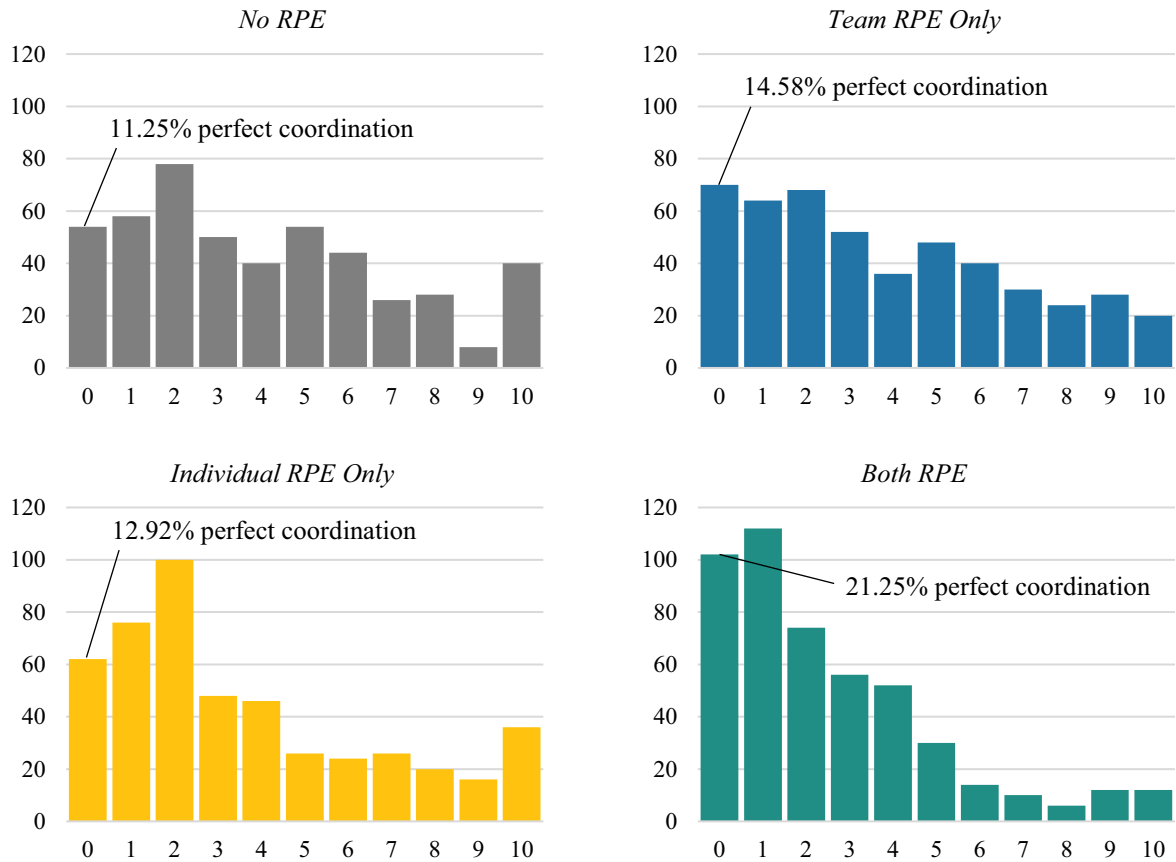
Notes. The x-axes in this figure represent the levels of net indirect actions within the possible range $[-10, 10]$. The y-axes represent the number of participant-period observations at each of those levels. Each graph reports the results for each condition separately: *No RPE*, *Team RPE Only*, *Individual RPE Only*, and *Both RPE*. Please see Table B.1 in Appendix B for further details about our variable definitions.

are present (coefficient = -1.29 , $t = -4.53$, $p < 0.01$). Additionally, panel D reports the results of two F -tests for the equality of variances in direct effort further validating that individual RPE significantly reduce the variance in actual direct effort—an alternative proxy for strategic uncertainty about direct effort—whether team RPE are absent ($p = 0.08$) or present ($p < 0.01$).

Hypothesis 3b predicts a null effect of individual RPE on strategic uncertainty about indirect actions. Examining the pattern of results in panel A of Table 5, univariate tests indicate a significant negative effect of individual RPE when team RPE are present, as *Uncertainty Indirect* is lower in the *Both RPE* condition compared with the *Team RPE Only* condition (difference = -0.79 , $t = -2.98$, $p < 0.01$). However, we observe no significant difference between the *Individual RPE Only* condition and the *No RPE* condition (difference = 0.22 , $t = 0.65$, $p = 0.52$). Recall that there is significant tension underlying Hypothesis 3b, especially when there are only weak incentives to cooperate (e.g., when team RPE are absent). That is, individual RPE reflect strong

incentives for negative indirect actions, which may more than offset the weak directional incentives to cooperate that are inherent in teams without any RPE. This appears consistent with our findings, as we observe a slight increase in *Uncertainty Indirect* between the *Individual RPE Only* and *No RPE* conditions.

Panels B and C report the results of our multivariate tests that further support these observed effects. Specifically, panel C confirms that the effect of individual RPE on *Uncertainty Indirect* is insignificant when team RPE are absent (coefficient = 0.47 , $t = 1.13$, $p = 0.26$), whereas this effect is negative and significant when team RPE are present (coefficient = -0.85 , $t = -2.79$, $p < 0.01$). Additionally, in untabulated F -tests, we observe that the use of either form of RPE in isolation increases the variance in actual net indirect actions compared with when no RPE are present ($p < 0.01$). However, as reported in panel D of Table 5, the presence of individual RPE significantly reduces this variance when both RPE are present (i.e., *Both RPE* versus *Team RPE Only*: $F = 1.98$, $p < 0.01$). Moreover, when both RPE are present, the variance in

Figure 2. (Color online) Frequency of Coordination Failure by Condition

Notes. The x-axes in this figure represent the degree of coordination failure, where coordination failure is defined as the absolute difference in a participant's own output and that of their team member, within the range [0,10]. Zero coordination failure reflects perfect coordination (i.e., full efficiency). The y-axes represent the number of participant-period observations at the specified degree of coordination failure. Each graph reports the results for each condition separately: *No RPE*, *Team RPE Only*, *Individual RPE Only*, and *Both RPE*. Please see Table B.1 in Appendix B for further details about our variable definitions.

Net Indirect Actions is at its lowest, falling even below the *No RPE* condition ($F = 1.44, p < 0.01$; untabulated).

4.4. Tests of Hypothesis 4a and Hypothesis 4b: Team Output

Our final hypothesis predicts that a reduction in strategic uncertainty about direct effort (Hypothesis 4a) and indirect actions (Hypothesis 4b) improves team output. To test these predictions, we regress *Team Output* on *Uncertainty_Direct* and *Uncertainty_Indirect* while controlling for *Direct Effort* and *Net Indirect Actions*. As shown in panel B of Table 6, we find that each component of strategic uncertainty has a significant negative effect on *Team Output* (both $p < 0.01$), providing support for our Hypothesis 4a and Hypothesis 4b.

Collectively, the predicted effects of Hypothesis 3 and Hypothesis 4 imply that individual RPE should have a beneficial impact on team output because of their reduction of strategic uncertainty. Specifically, a reduction in strategic uncertainty should mediate the positive effect of individual RPE on team output. To

explore this potential mediation, we first establish that individual RPE improve team output in panel A of Table 6. Although not formally hypothesized, we also find that individual RPE increase team output (*Individual RPE Only* versus *No RPE*: difference = 2.78, $t = 10.59, p < 0.01$; and *Both RPE* versus *Team RPE Only*: difference = 1.33, $t = 3.18, p < 0.01$).

Next, we examine the mediating effect of each component of strategic uncertainty on the increase in *Team Output* stemming from individual RPE.²² We conduct our mediation analysis using a structural equation model, consistent with Hayes (2022), with the following three regression models:

1. $Uncertainty_Direct = a_0 + a_1 Individual\ RPE$
2. $Uncertainty_Indirect = b_0 + b_1 Individual\ RPE$
3. $Team\ Output = c_0 + c_1 Individual\ RPE + c_2 Uncertainty_Direct + c_3 Uncertainty_Indirect + c_4 Direct\ Effort + c_5 Net\ Indirect\ Actions$

Each of these models include period fixed effects and clustered robust standard errors. As our hypotheses focus on the simple effects of individual RPE, we report

Table 4. Effects of the Form of RPE on *Uncertainty_Direct*

Panel A: Mean comparisons, univariate					
	<i>No RPE</i> (1)	<i>Team RPE Only</i> (2)	<i>Individual RPE Only</i> (3)	<i>Both RPE</i> (4)	
Mean (standard deviation)	3.88 (2.98)	3.64 (2.62)	3.12 (3.16)	2.57 (2.30)	
	Prediction	Difference	Standard error	<i>t</i> -statistic	<i>p</i> -value
Hypothesized comparisons					
(3) versus (1)	– (Hypothesis 3a)	–0.76	0.27	–2.81	<0.01
(4) versus (2)	– (Hypothesis 3a)	–1.07	0.24	–4.41	<0.01
Other comparisons					
(2) versus (1)		–0.24	0.26	–0.93	0.36
(4) versus (1)		–1.31	0.25	–5.23	<0.01
(3) versus (2)		–0.52	0.26	–1.97	0.05
(4) versus (3)		–0.55	0.26	–2.14	0.04
Panel B: Full sample, multivariate					
	Coefficient (standard error)		<i>t</i> -statistic	<i>p</i> -value	
<i>Team RPE</i>	–0.14 (0.31)		–0.44	0.66	
<i>Individual RPE</i>	–1.04 (0.35)		–2.97	<0.01	
<i>Team RPE</i> × <i>Individual RPE</i>	–0.33 (0.46)		–0.71	0.48	
Intercept	3.28 (0.35)		9.40	<0.01	
<i>n</i> = 1,920 (128 clusters), <i>F</i> (17, 1,903) = 4.22					
Panel C: Simple effects, multivariate					
	Prediction	Difference	Standard error	<i>t</i> -statistic	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> absent	– (Hypothesis 3a)	–1.11 (0.37)	–2.97	<0.01	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> present	– (Hypothesis 3a)	–1.29 (0.29)	–4.53	<0.01	
Panel D: <i>F</i> -tests for equality of variances in <i>Direct Effort</i>					
	Ratio		<i>F</i> -statistic	<i>p</i> -value	
<i>Individual RPE Only</i> (standard deviation = 3.19) versus <i>No RPE</i> (standard deviation = 3.46)	0.92		1.17	0.08	
<i>Both RPE</i> (standard deviation = 2.60) versus <i>Team RPE</i> <i>Only</i> (standard deviation = 3.21)	0.81		1.53	<0.01	

Notes. This table reports results related to *Uncertainty_Direct* that reflect the absolute value of the error in the participant's explicit predictions of their team member's direct effort, within the range [0,10]. All *p*-values reflect two-tailed tests, and all variables reflect those defined in Table B.1 in Appendix B. Panel A reports the univariate results of a series of *t*-tests of mean differences across conditions with unequal variances at the participant level (*n* = 32 per condition). Conditions are labeled (1) No RPE, (2) Team RPE Only, (3) Individual RPE Only, and (4) Both RPE. Hypothesized comparisons are annotated with directional predictions, and all other comparisons are reported for completeness. Panels B and C report the results of multivariate Tobit regressions with *Uncertainty_Direct* as the dependent variable. Each model includes period fixed effects and clusters robust standard errors at the participant level. Panel B includes the full sample (*n* = 1,920) with 335 left-censored and 101 right-censored observations. All results are robust to using OLS as detailed in the Online Appendix. Panel C reports the hypothesized simple effects from these multivariate tests. Panel D reports the results of traditional *F*-tests for equality of variances in actual effort (*Direct Effort*) across conditions, which are also robust to Levene's tests that relax the assumption of normality.

the results of this structural equation model analysis in Figure 3 both when team RPE are absent (panel A) and when team RPE are present (panel B).²³

As illustrated in Figure 3, we first confirm the predicted effects of Hypothesis 4 hold across both panels; that is, both components of strategic uncertainty (*Uncertainty_Direct* and *Uncertainty_Indirect*) significantly reduce Team Output (all *p* < 0.01). We then examine the simple effects of Individual RPE on each component of strategic uncertainty and test for mediation through their specific indirect effects on Team Output.²⁴ In panel

A, we find that Individual RPE significantly reduce strategic uncertainty about direct effort (*Uncertainty_Direct*) when team RPE are absent (coefficient = –0.76, *z* = –2.83, *p* < 0.01), resulting in a significant positive indirect effect on Team Output (indirect effect = 0.20, *z* = 3.52, *p* < 0.01). In contrast, we again find that Individual RPE do not significantly affect strategic uncertainty about indirect actions (*Uncertainty_Indirect*) when team RPE are absent (*p* = 0.51), thus having no significant indirect effect (*p* = 0.31). Although not hypothesized, we also test for significant differences in these indirect effects. We

Table 5. Effects of the Form of RPE on *Uncertainty_Indirect*

Panel A: Mean comparisons, univariate					
	<i>No RPE</i> (1)	<i>Team RPE Only</i> (2)	<i>Individual RPE Only</i> (3)	<i>Both RPE</i> (4)	
Mean (standard deviation)	2.98 (2.99)	2.98 (2.63)	3.20 (3.07)	2.19 (2.17)	
	Prediction	Difference	Standard error	<i>t</i> -statistic	<i>p</i> -value
Hypothesized comparisons					
(3) versus (1)	? (Hypothesis 3b)	0.22	0.34	0.65	0.52
(4) versus (2)	? (Hypothesis 3b)	−0.79	0.27	−2.98	<0.01
Other comparisons					
(2) versus (1)		0.01	0.33	0.02	0.99
(4) versus (1)		−0.79	0.28	−2.78	<0.01
(3) versus (2)		0.22	0.33	0.66	0.51
(4) versus (3)		−1.01	0.28	−3.57	<0.01
Panel B: Full sample, multivariate					
		Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
<i>Team RPE</i>		0.18 (0.40)	0.44	0.66	
<i>Individual RPE</i>		0.44 (0.41)	1.08	0.28	
<i>Team RPE</i> × <i>Individual RPE</i>		−1.29 (0.51)	−2.53	0.01	
Intercept		3.12 (0.37)	8.39	<0.01	
<i>n</i> = 1,920 (128 clusters), <i>F</i> (17, 1,903) = 3.54					
Panel C: Simple effects, multivariate					
	Prediction	Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> absent	? (Hypothesis 3b)	0.47 (0.42)	1.13	0.26	
Effect of <i>Individual RPE</i> if <i>Team RPE</i> present	? (Hypothesis 3b)	−0.85 (0.30)	−2.79	<0.01	
Panel D: <i>F</i> -tests for equality of variances in <i>Net Indirect Actions</i>					
		Ratio	<i>F</i> -statistic	<i>p</i> -value	
<i>Individual RPE Only</i> (standard deviation = 2.15) versus <i>No RPE</i> (standard deviation = 1.48)		1.45	0.47	<0.01	
<i>Both RPE</i> (standard deviation = 1.23) versus <i>Team RPE</i> <i>Only</i> (standard deviation = 1.73)		0.71	1.98	<0.01	

Notes. This table reports results related to *Uncertainty_Indirect* that are equal to the absolute value of the difference between each participant's implicit expectation of their team member's indirect actions and their team member's actual indirect actions within the range [0,20]. All *p*-values reflect two-tailed tests, and all variables reflect those defined in Table B.1 in Appendix B. Panel A reports the univariate results of a series of *t*-tests of mean differences across conditions with unequal variances at the participant level ($n = 32$ per condition). Conditions are labeled (1) No RPE, (2) Team RPE Only, (3) Individual RPE Only, and (4) Both RPE. Hypothesized comparisons are annotated with the null prediction, and all other comparisons are reported for completeness. Panels B and C report the results of multivariate Tobit regressions with *Uncertainty_Indirect* as the dependent variable. Each model includes period fixed effects and clusters robust standard errors at the participant level. Panel B includes the full sample ($n = 1,920$) with 335 left-censored and no right-censored observations. All results are robust to using OLS as detailed in the Online Appendix. Panel C reports the hypothesized simple effects from these multivariate tests. Panel D reports the results of traditional *F*-tests for equality of variances in actual cooperation (*Net Indirect Actions*) across conditions, which are also robust to Levene's tests that relax the assumption of normality.

find that the indirect effect via *Uncertainty_Direct* is significantly greater than via *Uncertainty_Indirect* when team RPE are absent (difference = 0.22, $z = 3.71$, $p < 0.01$).

In panel B, we find similar results for mediation through strategic uncertainty about direct effort when team RPE are present. However, we now also observe mediation through *Uncertainty_Indirect* as well. That is, when team RPE are present, *Individual RPE* significantly reduce *Uncertainty_Indirect* (coefficient = −0.79, $z = -3.00$,

$p < 0.01$), resulting in a significant positive indirect effect on *Team Output* (indirect effect = 0.19, $z = 4.07$, $p < 0.01$). Therefore, it appears that the positive effect of *Individual RPE* on *Team Output* is simultaneously mediated by both components of strategic uncertainty when team RPE are present. Comparing these indirect effects, we again observe that the indirect effect through *Uncertainty_Direct* is significantly greater than through *Uncertainty_Indirect* in this sample (difference = 0.22, $z = 2.61$, $p < 0.01$).

Table 6. Effects of Strategic Uncertainty on *Team Output*

Panel A: Mean comparisons, univariate				
	No RPE (1)	Team RPE Only (2)	Individual RPE Only (3)	Both RPE (4)
Mean (standard deviation)	22.22 (2.54)	24.87 (2.97)	25.00 (3.07)	26.20 (2.76)
	Difference	Standard error	<i>t</i> -statistic	<i>p</i> -value
(3) versus (1)	2.78	0.26	10.59	<0.01
(4) versus (2)	1.33	0.42	3.18	<0.01
(2) versus (1)	2.65	0.36	7.42	<0.01
(4) versus (1)	3.98	0.33	11.88	<0.01
(3) versus (2)	0.13	0.36	0.36	0.72
(4) versus (3)	1.20	0.34	3.53	<0.01
	Prediction	Coefficient (standard error)	<i>t</i> -statistic	<i>p</i> -value
<i>Uncertainty_Direct</i>	– (Hypothesis 4a)	–0.43 (0.05)	–8.08	<0.01
<i>Uncertainty_Indirect</i>	– (Hypothesis 4b)	–0.20 (0.03)	–5.93	<0.01
<i>Direct Effort</i>		0.68 (0.35)	19.70	<0.01
<i>Net Indirect Actions</i>		0.57 (0.44)	13.06	<0.01
Intercept		8.11 (1.03)	7.92	<0.01
<i>n</i> = 1,920 (128 clusters), <i>F</i> (18, 1,902) = 53.88				

Notes. This table reports results related to *Team Output* that reflect total team output, that is, the lower of the team members' individual outputs within the range [20,30]. All *p*-values reflect two-tailed tests, and all variables reflect those defined in Table B.1 in Appendix B. Panel A reports the univariate results of a series of *t*-tests of mean differences across conditions with unequal variances at the participant level (*n* = 32 per condition). Conditions are labeled (1) *No RPE*, (2) *Team RPE Only*, (3) *Individual RPE Only*, and (4) *Both RPE*. Panel B reports the results of our hypothesis tests using a multivariate Tobit regression with *Team Output* as the dependent variable. This model includes the full sample (*n* = 1,920) with 344 left-censored and 118 right-censored observations and includes period fixed effects and clusters robust standard errors at the participant level. Results are robust to using OLS as detailed in the Online Appendix.

4.5. Supplemental Analysis: Time Trends

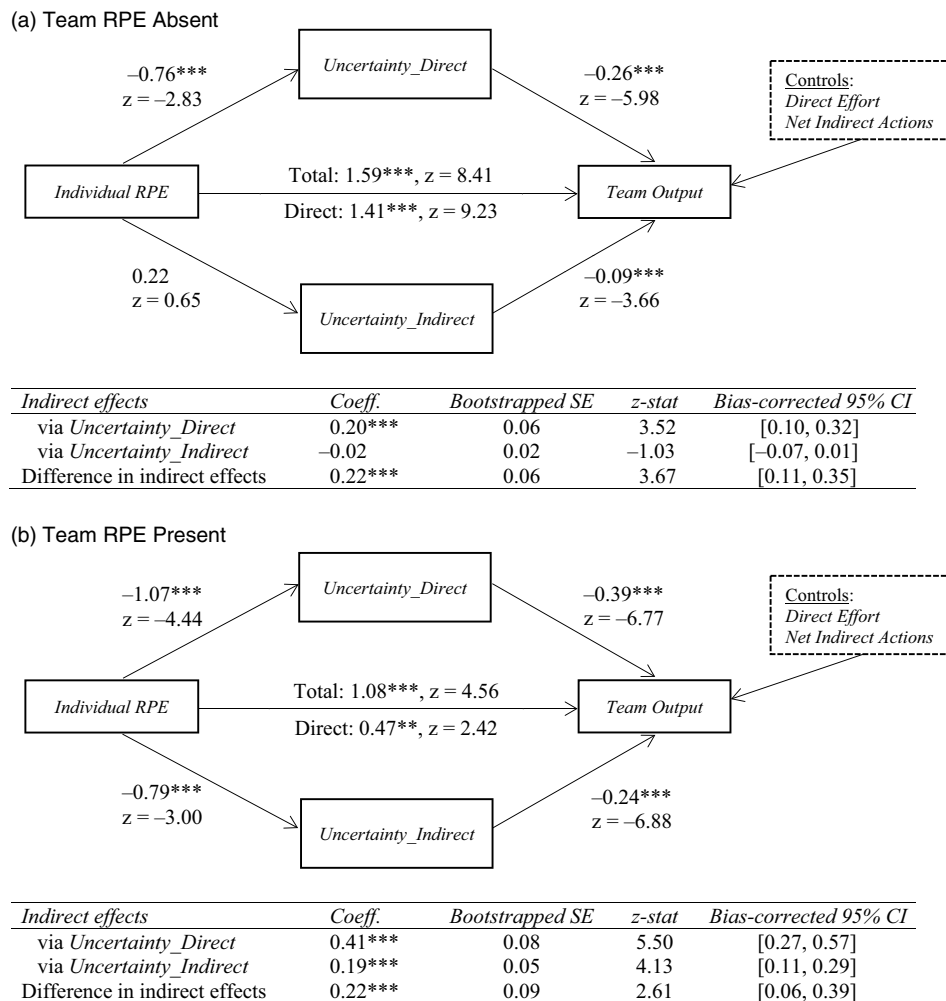
In addition to a comprehensive set of robustness tests reported in the Online Appendix, we also conduct several supplemental analyses to explore general performance trends in *Team Output* over time. Whereas participants are randomly rematched with new team members each period and the results of the tests of our hypotheses are robust across all periods, we identify several interesting performance trends more generally over time. For this analysis, we compare mean *Team Output* between the first (periods 1–7) and second (periods 8–15) halves of the experiment for each condition separately. A series of untabulated *t*-tests reveals a significant negative trend in *Team Output* over time in the *No RPE* condition (difference = –0.76, *t* = –3.28, *p* < 0.01) that is consistent with prior research using similar coordination settings (e.g., Van Huyck et al. 1990, Riechmann and Wiemann 2008). Interestingly, the use of team RPE appears to mitigate this downward trend, as we observe (1) higher *Team Output* in the *Team RPE Only* condition compared with the *No RPE* condition each period (all untabulated *p* < 0.10), and (2) insignificant differences in *Team Output* within the *Team RPE Only* condition over time (difference = –0.13, *t* = –0.46, *p* = 0.65; untabulated).

Interestingly, we also observe a significant improvement in *Team Output* over time when individual RPE

are present (*Individual RPE Only*: difference = 0.92, *t* = 3.30, *p* < 0.01; and *Both RPE*: difference = 1.41, *t* = 5.73, *p* < 0.01; untabulated). More closely examining these two conditions, we observe similar positive trends in *Direct Effort* between the first and second halves of the experiment, as well as decreasing *Uncertainty_Direct* and *Uncertainty_Indirect* (all untabulated *p* < 0.01). In further tests, we find consistent time trend results when comparing the first five periods to the last five periods as well. These general trends across conditions lend further support for the theory underlying our hypotheses.

5. Conclusion

Overall, our study highlights an important novel benefit to using individual RPE to motivate interdependent teams where strategic uncertainty is a critical deterrent to team success. We find that individual RPE reduce employees' strategic uncertainty about their team members' direct effort. Similarly, we also find that individual RPE reduce employees' strategic uncertainty about their team members' indirect actions when teams are strongly motivated by cooperative incentives. Furthermore, reducing each of these distinct components of strategic uncertainty ultimately leads to better coordination with higher levels of coordinated effort and, in doing so, improves team outcomes.

Figure 3. Mediation Analysis: Components of Strategic Uncertainty

Notes. This figure reports the results of a structural equation model, whereby the positive effect of *Individual RPE* on *Team Output* is mediated by a reduction in strategic uncertainty (*Uncertainty_Direct* and *Uncertainty_Indirect*). (a) The results of this mediation when team RPE is absent ($n = 960$). (b) The results when team RPE are present ($n = 960$). All models include period fixed effects and cluster the robust standard errors at the participant level. All indirect effects reflect bootstrapped standard errors with 5,000 replications. For completeness, please refer to the Online Appendix, which reports all direct, indirect, and total effects for all variables.

Our study makes several important contributions. Whereas researchers and commentators often focus on the negative effects of competitive incentives on team outcomes, individual RPE remain ubiquitous in practice. Our results help bridge this disparity by providing key insights into the positive effects of individual RPE and documenting how firms can benefit from using competitive incentives in cooperative team settings. Our results highlight how the benefits of reducing strategic uncertainty can dominate the negative effects of competition on cooperation and result in improved coordination and team output.

Furthermore, we provide a richer understanding of the distinct components of strategic uncertainty commonly present in practice (i.e., through direct effort and indirect actions). If firms prioritize improving direct effort and/or reducing negative indirect actions

(e.g., sabotage), then the use of individual RPE within interdependent teams may improve coordination and team output. However, when positive indirect actions (e.g., knowledge sharing) are integral to a firm, our results would still caution against the use of individual RPE. Specifically, we find that when teams are strongly motivated by cooperative incentives, individual RPE reduce indirect actions in either direction and effectively reduce helping. Therefore, our study importantly suggests that managers consider the trade-offs between these factors based on the relative importance of direct effort and indirect actions to their firm.

As with all studies, our study is subject to limitations that highlight opportunities for future research. We carefully introduce important elements from practice through the simultaneous ability of employees to both help and hurt their team members when working in

interdependent teams. Yet future research can consider settings with differing forms of helping and hurting actions, such as punishment, or settings with asymmetric costs for helping and hurting. We also chose to randomly rematch participants between periods in a setting without communication to provide a clean test of theory regarding the distinct effects of individual RPE on strategic uncertainty. However, prior literature shows that cultural controls, reputation effects, and social norm development can also have a significant impact on employee behavior (Merchant and Van der Stede 2024). The moderating effects of these aspects on our theory appear to be a fruitful direction for future research extending our study. For example, future research could examine the effects of reputation on strategic uncertainty in isolation and in conjunction with various levels of RPE. We also use a setting with small teams of homogeneous employees; however, future research can examine whether the effects of RPE differ in settings where teams are less homogeneous or larger in size. Finally, we only use one form of reward linked to individual RPE (i.e., a cash bonus). We encourage future research to explore whether other forms of reward or penalty (e.g., promotions, dismissals) may yield distinct effects. Moreover, future research could explore the

interaction between individual RPE and incentive intensity, as it is possible that more intense incentives could induce giving-up behavior in some employees that may moderate the effectiveness of this incentive at reducing strategic uncertainty, particularly in teams with vast heterogeneity among team member abilities.

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Appendix A. Payoff Table

Table A.1. Payoff Table

Base Pay = Minimum output – (0.50 × Initial choice)												
		Minimum output between you and your team member										
		30	29	28	27	26	25	24	23	22	21	20
Initial choice	30	15.00	14.00	13.00	12.00	11.00	10.00	9.00	8.00	7.00	6.00	5.00
	29	15.50	14.50	13.50	12.50	11.50	10.50	9.50	8.50	7.50	6.50	5.50
	28	16.00	15.00	14.00	13.00	12.00	11.00	10.00	9.00	8.00	7.00	6.00
	27	16.50	15.50	14.50	13.50	12.50	11.50	10.50	9.50	8.50	7.50	6.50
	26	17.00	16.00	15.00	14.00	13.00	12.00	11.00	10.00	9.00	8.00	7.00
	25	17.50	16.50	15.50	14.50	13.50	12.50	11.50	10.50	9.50	8.50	7.50
	24	18.00	17.00	16.00	15.00	14.00	13.00	12.00	11.00	10.00	9.00	8.00
	23	18.50	17.50	16.50	15.50	14.50	13.50	12.50	11.50	10.50	9.50	8.50
	22	19.00	18.00	17.00	16.00	15.00	14.00	13.00	12.00	11.00	10.00	9.00
	21	19.50	18.50	17.50	16.50	15.50	14.50	13.50	12.50	11.50	10.50	9.50
	20	20.00	19.00	18.00	17.00	16.00	15.00	14.00	13.00	12.00	11.00	10.00

Remember:

If you changed your team member’s choice, deduct your change fee from the base pay shown above to calculate your Net Pay.
Net Pay = Base Pay – Change fee

Notes. This table reflects the payoff matrix provided to help participants calculate their base pay. Rows reflect the participant’s own “initial choice,” whereas the columns reflect the “minimum output” each period. Note that “base pay” excludes consideration of any bonuses. Reference to any bonuses was restricted to the relevant conditions. For example, in the *Team RPE Only* and *Individual RPE Only* conditions, the formula for net pay was stated as: Net Pay = Base Pay – Change Fee + Bonus. In the *Both RPE* condition, the formula for net pay was stated as: Net Pay = Base Pay – Change Fee + Individual Bonus + Team Bonus.

Several examples illustrating the functionality of the payoff table were provided to participants. Participants first considered an example with no indirect actions, when a team member’s initial choice is 22 and their own initial choice is 25. Net pay would be \$9.50, as the minimum output would be limited to 22, despite their own choice of 25. Follow-up examples illustrate how positive and negative indirect actions would affect their pay, considering the change fee of \$0.50 per point; for example, increasing their team member’s output by four points would raise the minimum to 25, resulting in net pay of \$10.50 (\$12.50 base pay – \$2.00 change fee).

Appendix B. Variable Descriptions

Table B.1. Variable Descriptions

Variable	Description
Independent variables	
<i>No RPE</i>	Indicator for the condition where no RPE bonuses are present.
<i>Team RPE Only</i>	Indicator for the condition where a single bonus is paid to the team with the higher team output, comparing each team to one other team.
<i>Individual RPE Only</i>	Indicator for the condition where a single bonus is paid to the individual team member who produced the highest individual output within their team.
<i>Both RPE</i>	Indicator for the condition where both the team RPE and individual RPE bonuses are present.
Dependent variables	
<i>Direct Effort</i>	Each participant's chosen direct effort within the range [20,30].
<i>Net Indirect Actions</i>	The amount by which each participant chose to change their team member's output within the range [−10,10].
<i>Uncertainty_Direct</i>	The absolute value of the error in the participant's explicit predictions of their team member's direct effort, within the range [0,10].
<i>Uncertainty_Indirect</i>	The absolute value of the difference in the participant's implicit expectation of their team member's indirect actions and their team member's actual indirect actions, within the range [0,20].
<i>Coordination Failure</i>	The absolute difference in a participant's own output and that of their team member, within the range [0,10].
<i>Team Output</i>	The lower of the team members' individual outputs, within the range [20,30].

Endnotes

¹ For example, the automotive manufacturing industry is heavily reliant on complex, sequential assembly lines that are highly sensitive to weak links that create bottlenecks (Lai et al. 2021). Similarly, an airline's ability to deliver passengers on time is most strongly affected by its underperformers, as shortcomings within various crews (e.g., fueling, maintenance, baggage handling) harm the other team members' efforts to perform (Katz and Sider 2022).

² Some research refers to additive groups as "teams" (e.g., Arnold et al. 2018), to interdependent teams as "groups" (e.g., Chen et al. 2012), or to both additive groups and teams as "teams" (e.g., Arnold and Tafov 2019). We use the term "teams" to specifically describe interdependent groups.

³ Consistent with research using minimum-effort coordination games, we examine a setting with homogenous employees. As the primary focus of our study is not on the effects of differential ability within teams, this design choice provides a clean test of theory, although differential ability likely contributes to increased strategic uncertainty.

⁴ Firms in a variety of industries, such as software development, film production, manufacturing, and construction, commonly structure their operations as interdependent teams of largely homogenous employees. For example, Kelly et al. (2026) study a manufacturing company whose production process employs three-person interdependent teams of homogenous employees that rotate between stations at fixed intervals. Intrateam individual RPE are highly relevant in such settings, as they control for common uncertainty at the most granular level.

⁵ Research on cooperation commonly uses voluntary-contribution games (e.g., Chaudhuri 2011, Cartwright and Lovett 2014) or trust games (Dal Bó and Fréchette 2011, Breitmoser 2015) that focus on either individual or team incentives and largely ignore indirect actions. Moreover, although voluntary-contribution, or "public good," games involve cooperation, they do not reflect interdependent teams, as an individual's marginal return is unaffected by the actions of others. As such, these settings do not require coordination.

⁶ The reference group for individual RPE is often all employees at the same level within a department, division, or firm (e.g., promotion decisions). RPE can remove common uncertainty from these evaluations by making comparisons using a similar environment, such as within a team of employees (Matsumura and Shin 2006). Given our theory focuses on competitive incentives within interdependent teams, we examine individual RPE that compare the performance of members of the same team. Other forms of individual RPE (e.g., interlevel or across teams) require additional assumptions regarding the exact composition of teams and/or job functions that are not fundamental to our research question. Therefore, such tests are outside the scope of this study but provide opportunities for future research.

⁷ In practice, the optimal level of effort is established based on the expected benefit and cost of that effort. Providing higher effort, which presents a certain higher cost, is risky in that the benefit of that effort may not realize if employees do not accurately predict their team members' actions and, therefore, the value of their own effort.

⁸ The level of interdependence in this setting is consistent with general interdependency. We expect an employee to provide effort such that the marginal benefit, which depends on a team member's choices, is at least equal to the marginal cost. If an employee provides extra unmatched effort (e.g., in expectation of higher effort from one's team member), this extra effort is a net loss to the employee, as the marginal cost exceeds the marginal benefit.

⁹ The primary focus of our study is on strategic uncertainty. Firms can also use RPE to eliminate common uncertainty among employees (Dye 1984). For simplicity, we do not examine common uncertainty in our setting, as it is not central to our theory involving strategic uncertainty. However, our setting does reflect firms' contracting on observable output. As such, our design generalizes to a setting where common uncertainty exists.

¹⁰ This study was reviewed and approved by the relevant institution's institutional review board (IRB).

¹¹ An important distinction of our study compared to other minimum-effort coordination games is the consideration of indirect

actions. Participants could expend costly effort (i.e., \$0.50 per point) to change their team member's output. For simplicity, the payoffs presented in Table A.1 in Appendix A reflect base pay determined based on direct effort (referred to as their "initial choice") and minimum team output. Net pay is equal to base pay less the cost of any indirect actions. Several examples illustrating the functionality of the payoff table were provided to participants in the instructions that were read aloud by one of the researchers at the start of each session. Also, participants completed five practice periods at the start of the experiment to familiarize themselves with the payoffs before the live periods.

¹² Research suggests that relative performance information (RPI) can affect behavior without affecting pay (Hannan et al. 2013). Therefore, we hold constant the amount and form of RPI by informing all participants of the output of their team member and a peer team each period in all conditions. We varied only the reference used for evaluations.

¹³ The guided language in the *Team RPE Only* condition focused on team output; that is, "I want to [increase/decrease] the chance my team's minimum output is greater than the peer team" or "I do not want to make any changes." The guided language in the *Individual RPE Only* condition focused on individual output; that is, "I want to [increase/decrease] the chance my team member's output is greater than my own" or "I do not want to make any changes." We displayed the guided language for both the *Team RPE Only* and *Individual RPE Only* conditions in the *Both RPE* condition.

¹⁴ All reported results in Section 4 are robust to excluding the guided-language periods.

¹⁵ Participants understood the incentives captured in these equilibria as they agreed with the statement "I always tried to match what I expected my team member did" on a seven-point Likert scale, both with (mean = 4.58) and without (mean = 4.59) individual RPE present (each mean significantly higher than the midpoint at $p \leq 0.02$). However, they agreed more (less) with the statement "I always tried to pick a number higher (lower) than my team member" when individual RPE were present (means of 5.80 versus 3.94 for "pick higher" and 2.34 versus 3.36 for "pick lower"; each comparison significantly different at $p < 0.01$). Therefore, whereas all participants appeared concerned with at least matching their team member's output, participants subject to individual RPE were significantly more concerned with exceeding it.

¹⁶ In untabulated tests, we confirm that our results involving the components of strategic uncertainty are robust in sign, significance, and inference to our general proxy of strategic uncertainty (*Coordination Failure*). We also construct an alternative specification of *Uncertainty Indirect* using the residual term when regressing the level of coordination failure (*Coordination Failure*) on strategic uncertainty about direct effort (*Uncertainty Direct*) each period (Chen et al. 2022). In untabulated tests, we verify that all our results are robust in sign, significance, and inference to using this alternative specification of strategic uncertainty about indirect actions.

¹⁷ Tobit regressions provide the most appropriate inferences when dealing with data subject to mechanical truncations. However, we also confirm our results are also qualitatively and statistically robust when using ordinary least squares (OLS).

¹⁸ Unless otherwise noted, all p -values are reported as two-tailed.

¹⁹ As the interaction in panel B of Table 2 is significantly negative, we perform an untabulated joint F -test (*Individual RPE* + (*Team RPE* $\times$ *Individual RPE*)), verifying that the combined effect is significant ($F(2, 1,903) = 33.00, p < 0.01$).

²⁰ In Figure 1, we also observe some minor negative indirect actions (i.e., *Hurting*) in the *No RPE* (mean = -0.08) and *Team RPE Only* (mean = -0.09) conditions despite a lack of individual-level incentives to do so. To provide assurance that these choices do not reflect

participants' misunderstanding and/or affect our results, we conduct several robustness tests for these two conditions. First, we confirm there is no significant difference in *Hurting* in the guided-language periods compared to other periods when individual RPE are absent ($t = 0.82, p = 0.41$), indicating participants understood the implications of their choices. Furthermore, we observe no other clear pattern across periods, indicating *Hurting* did not change substantially over time and is not exclusive to the first few periods. Second, we confirm that this behavior is not driven by just one or a few participants. Specifically, 18 unique participants engaged in hurting behavior when individual RPE were absent, 11 of which chose to do so more than once. We confirm that all of our results are robust in sign, significance, and inference when excluding these 18 participants.

²¹ We define perfect coordination as participants perfectly coordinating at the same output level, minimizing effort inefficiency. This definition does not mean that participants coordinated on the maximum output level of 30.

²² Our results are also robust when replacing the specific components of strategic uncertainty (i.e., *Uncertainty Direct* and *Uncertainty Indirect*) with the single broader measure of overall strategic uncertainty (i.e., *Coordination Failure*) as the mediator. Moreover, while *Team Output* is the same for both team members, team members change each period, and our control variables are all measured at the participant level. Therefore, our mediation tests retain all individual observations. As a more conservative test, we also conduct all analyses using participant-level averages. We note that all of our results are robust in sign, significance, and inference to our reported tests.

²³ See the Online Appendix for further details regarding our mediation analysis, wherein we comprehensively disclose all direct, indirect, and total effects for all variables (including controls) across each subsample and the full sample.

²⁴ We test for the specific indirect effects of individual RPE on team output via each component of strategic uncertainty (*Uncertainty Direct* and *Uncertainty Indirect*) separately, using the product of coefficients method (i.e., $a_1 \times c_2$ and $b_1 \times c_3$ based on the models defined above), bootstrapped with 5,000 replications.

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